RoadLens pilot run: results

Run on a Colab T4 GPU on 15 October 2026 with **RoadLens_standalone.ipynb**, profile **pilot**. Every number below was computed from the evaluation files saved in *runs_pilot-20261015T180640Z-1-001.zip*; no model was re-run for this report.

Setup in one paragraph

German traffic signs (GTSRB, 43 sign types) at 64 x 64 pixels. Models are trained on clean photos and tested on photos with 14 kinds of damage they never saw during training, each at 5 strengths (the strengthened damage settings of 15 Oct 2026). Belgian signs (BTSD) test transfer to another country. Pilot profile: **one seed (0)**, training shortened to 40% of the epochs, and the **first 2,000 of the 12,630 test photos**. 13 networks were trained in 64 minutes of GPU time; 5 ensembles reuse them without training.

Main findings

• **Clean photos are solved.** Every model gets 99.0-99.6% on the clean test photos. The differences between models only show up on damaged photos.

• **The proposed training works.** ResNet-18 on damaged photos goes from **73.0% to 86.8%** (+13.9 points) and its mCE from 280 to 123. DeiT-S goes from 86.7% to 90.3%.

• **Combining two different models helps.** ResNet-18 + DeiT-S together reach about 90.5-90.8%, better than either alone. The learned gate (90.54%) did **not** beat a plain 50/50 average (90.82%); the gap is too small to matter in a single-seed run.

• **Best model: the average over 'view experts' (92.92%, mCE 64.9).** Showing each model a median-filtered and a blurred copy of the photo as well fixes most noise damage (impulse noise 80% -> 98%). The v2 stacked gate reaches 92.37% (mCE 78.7).

• **Final architecture (v2 stacked view-expert gate): 99.20% clean, 92.37% on damaged photos, 83.69% at the strongest damage (S5), mCE 78.7**; after temperature scaling its confidence error is 0.48% (clean). Full details on the next two pages.

• **Fog is unsolved.** Every model stays near 64-74% on fog; no method moved it.

• **Test-time adaptation fails at batch size 1** (ResNet-18 proposed: 72% -> 29% with TENT) and helps at 16+ (+5 to +14 points for single ResNets). A small 'source prior' fixes the batch-size-1 collapse.

• **Confidence is honest after one fix.** All models are under-confident (ECE about 9-12%) because of label smoothing; temperature scaling cut the v2 model's clean ECE from 9.0% to 0.5%.

How much to trust these numbers: one seed and 2,000 test photos. Differences below about one percentage point (for example gate vs. average) can come from chance alone. The full run (3 seeds, all 12,630 photos, full training length) is needed before claiming them.

The final architecture: v2 stacked view-expert gate

ResNet-18 and DeiT-S, both trained with the proposed method and with instance input normalisation. Each also looks at a median-filtered and a blurred copy of the photo (6 predictions in all). A small gate (998 parameters), fitted on held-out validation photos, combines them; temperature scaling (T = 0.60) then makes the confidence honest without changing any answer.

Clean accuracy	Damaged accuracy (14 x 5 sets)	Severity 5 accuracy	mCE (SmallCNN = 100)	Belgian signs, damaged	Confidence error (ECE, clean / damaged)
99.20%	**92.37%**	**83.69%**	**78.7**	**90.52%**	**0.48% / 4.11%**

Accuracy of the final architecture on all 14 damages at every strength

Damage	Family	S1	S2	S3	S4	S5	Mean	ResNet-18 baseline mean	Gain
gaussian noise	noise	96.2	92.7	85.9	77.2	65.0	**83.4**	31.7	+51.7
impulse noise	noise	98.2	96.9	96.8	94.8	91.7	**95.7**	26.6	+69.1
shot noise	noise	97.9	96.8	92.8	83.1	74.3	**89.0**	32.6	+56.4
motion blur	blur	99.0	98.8	98.2	97.2	96.1	**97.9**	96.0	+1.8
zoom blur	blur	99.2	99.2	99.0	98.6	97.9	**98.8**	95.8	+3.0
defocus blur	blur	99.2	99.0	98.9	98.3	96.6	**98.4**	96.8	+1.6
snow	weather	98.3	96.3	92.2	86.5	93.8	**93.4**	74.8	+18.6
fog	weather	91.4	85.3	76.3	61.3	**34.3**	**69.7**	67.5	+2.3
frost	weather	96.5	93.3	89.9	84.0	78.8	**88.5**	69.9	+18.6
brightness	weather	99.0	98.8	98.2	97.3	94.6	**97.6**	95.0	+2.5
contrast	digital	99.2	99.2	99.2	99.2	98.6	**99.1**	83.9	+15.1
pixelate	digital	99.2	99.2	99.0	98.9	97.9	**98.8**	87.7	+11.1
elastic transform	digital	99.2	98.3	95.1	86.6	69.8	**89.8**	80.4	+9.3
jpeg compression	digital	98.9	98.3	96.7	90.0	82.2	**93.2**	83.0	+10.3
All 14 (mean)		**98.0**	**96.6**	**94.2**	**89.5**	**83.7**	**92.4**	**73.0**	**+19.4**

Clean test photos: 99.20%. Red: below 60%. S1 = lightest, S5 = strongest damage. 2,000 test photos per cell, seed 0.

- Near-perfect (97-99%) on all blur types, brightness, contrast and pixelation, even at S5.
- Weak spots at S5: **fog (34%)**, gaussian noise (65%), elastic (70%) and shot noise (74%).

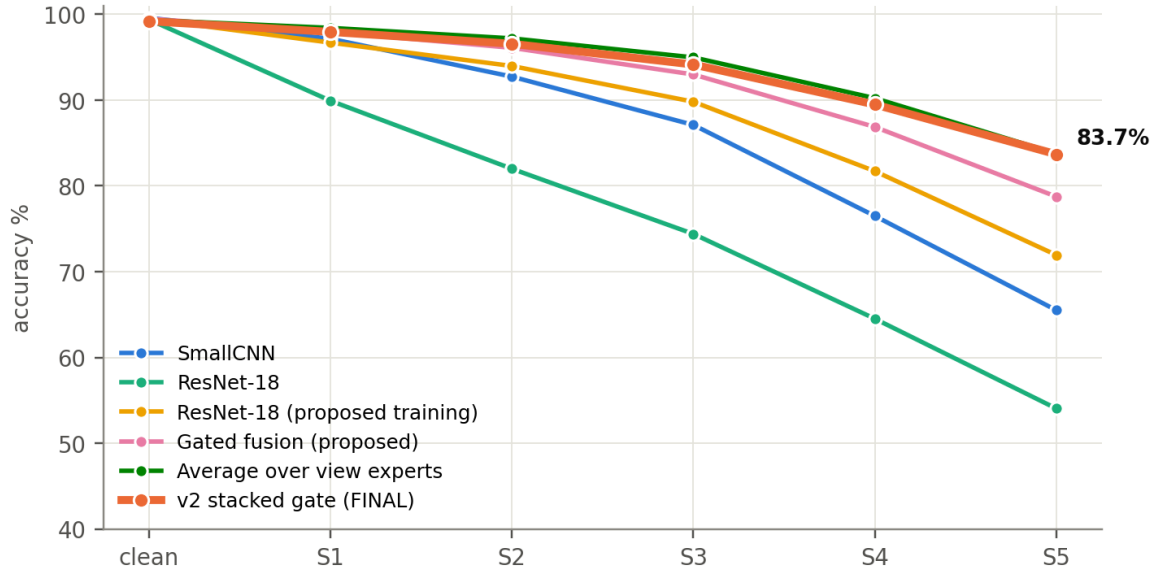
Accuracy at each damage strength, all key models

Model	Clean	S1	S2	S3	S4	S5	Mean of 70 damaged sets
SmallCNN	**99.60**	97.16	92.74	87.11	76.50	65.51	83.80
ResNet-18	99.40	89.93	82.00	74.40	64.51	54.05	72.98
DeiT-S	99.05	96.32	93.29	89.26	81.65	72.82	86.67
ResNet-18 (proposed training)	99.45	96.71	93.98	89.81	81.72	71.94	86.83
DeiT-S (proposed training)	99.50	97.88	95.99	92.79	86.63	78.39	90.34
Average ensemble	99.50	98.13	96.37	93.34	87.25	79.00	90.82
Gated fusion (proposed)	99.50	97.97	96.10	92.99	86.87	78.75	90.54
Average over view experts	99.35	**98.42**	**97.20**	**94.99**	90.23	83.77	92.92
View experts (instance norm)	99.30	98.21	96.98	94.68	**90.25**	**84.81**	**92.98**
v2 stacked gate (FINAL)	**99.20**	**97.96**	**96.57**	**94.16**	**89.50**	**83.69**	**92.37**

Accuracy %, mean of the 14 damages at each strength. Bold: best in the column. Orange row: the final architecture.

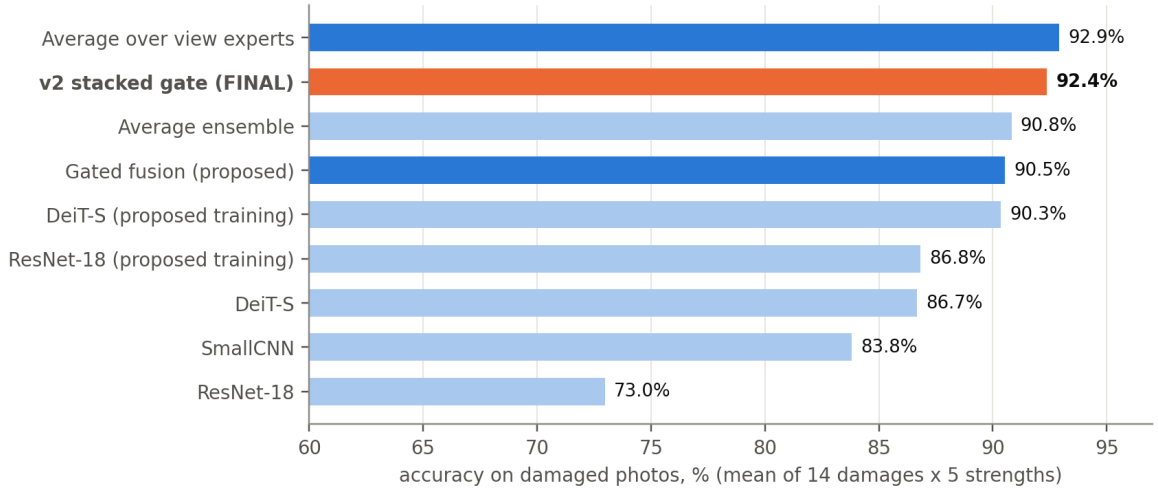
• The final architecture loses 15.5 points from clean to S5; the ResNet-18 baseline loses 45.4.

• On this single-seed pilot the plain averages over view experts are about 0.5-1.1 points higher than the final architecture at every strength. That gap is within chance for one seed and 2,000 photos; the full run decides it. The final architecture has the lowest confidence error of the three (damaged ECE 9.9% vs. 11.7-12.6% before calibration, section 9).



Orange line: the final architecture.

1. Damaged photos: the main result (Experiment 2)



Orange: the final architecture. Dark blue: the other combined models (v1 gated fusion, view experts). Mean accuracy over the 14 damages x 5 strengths, 2,000 photos each.

Model	Clean acc. %	Damaged acc. %	mCE	Rel. mCE	Damaged F1 %	Damaged ECE %
SmallCNN	99.60	83.80	100.0	100.0	82.71	17.98
ResNet-18	99.40	72.98	279.7	360.4	69.86	11.05
DeiT-S	99.05	86.67	129.8	125.3	84.63	8.01
ResNet-18 (proposed training)	99.45	86.83	123.4	136.8	85.08	9.82
DeiT-S (proposed training)	99.50	90.34	78.5	74.7	88.40	8.71
Average ensemble	99.50	90.82	73.5	68.3	89.11	11.29
Confidence-weighted ensemble	99.50	90.80	73.3	68.0	89.08	10.11
Gated fusion (proposed)	**99.50**	**90.54**	**76.4**	**72.1**	**88.65**	**9.11**
v2 stacked gate (FINAL)	**99.20**	**92.37**	**78.7**	**63.7**	**90.57**	**9.90**

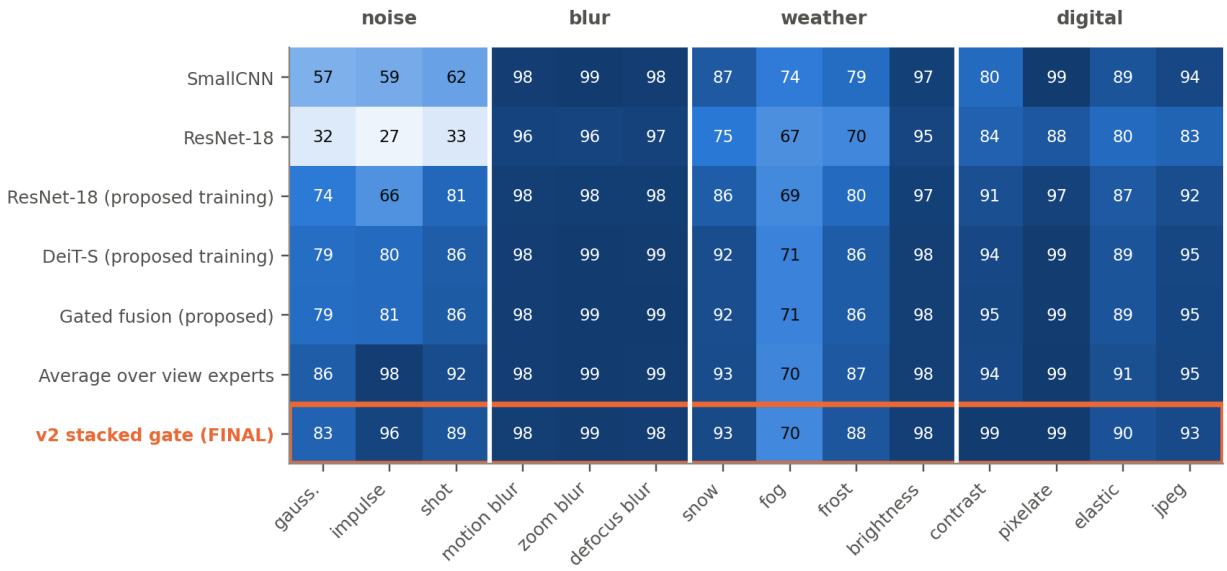
mCE: errors on damaged photos relative to SmallCNN (= 100), averaged over damages; lower is better. Relative mCE counts only the *extra* errors caused by the damage. ECE: gap between confidence and accuracy; lower is better.

What it means

• The small SmallCNN, trained from zero at 64 px, is surprisingly robust (83.8%), more than the ImageNet ResNet-18 baseline (73.0%). The ResNet baseline is very weak on noise (30% vs. 59%), which is why its mCE is 280.

• Only the proposed DeiT-S and the ResNet-18 + DeiT-S combinations make fewer damage errors than SmallCNN (mCE below 100). ResNet-18 with the proposed training has a higher mean accuracy than SmallCNN (86.8% vs. 83.8%) but still an mCE of 123, because mCE weighs every damage equally and SmallCNN is nearly perfect on blur.

2. Which damage hurts, and how much



Accuracy % per damage type (mean of the 5 strengths). Darker = better.

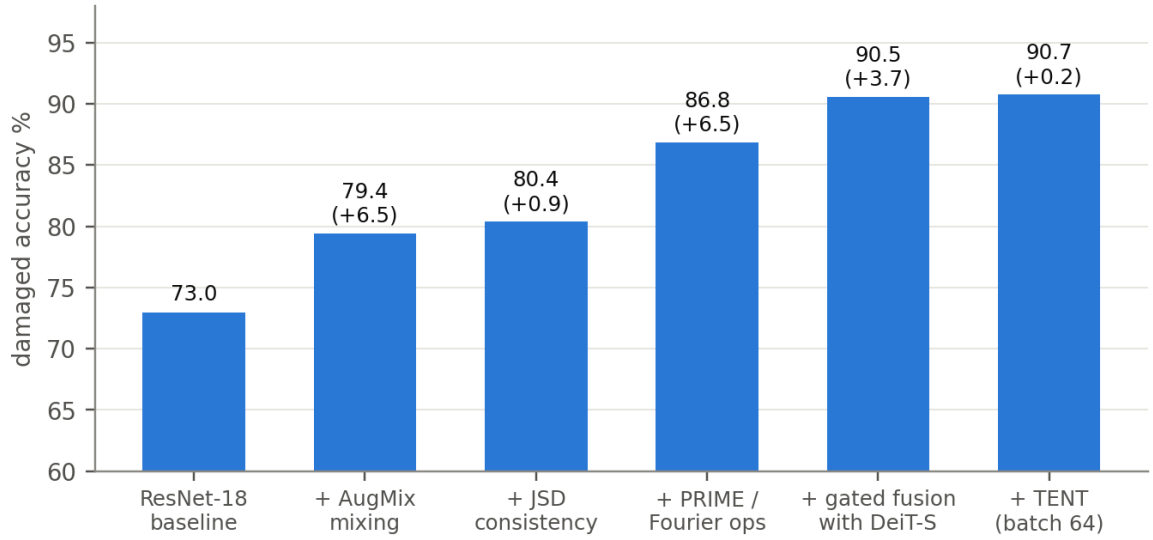
• **Noise** is the biggest weakness of the plain models (ResNet-18 30%) and the biggest win of the method: proposed training 74%, view experts 92%.

• **Blur** is nearly harmless at 64 px for every model (96-99%).

• **Fog** is the hardest single damage (64-74% for all models) and is the clearest target for future work.

• **Contrast** is fixed almost completely by instance input normalisation (ResNet-18 91% -> 99%, see section 9).

3. What each part adds (Experiment 3, ablation)



Each bar adds one piece to the previous one; the number in brackets is the change. mCE goes 280 -> 165 -> 166 -> 123 -> 76 -> 73.

• Mixing and the **Fourier/PRIME-style changes** are the two biggest steps (about +6.5 points each), then adding the second model (+3.7). The JSD 'same answer' rule adds only +0.9 here.

• TENT on top of the fusion adds just +0.2 points: in the fusion only the ResNet branch has BatchNorm layers to adapt (DeiT-S uses LayerNorm), and both branches are frozen behind the gate.

4. Clean photos (Experiment 1)

Model	Train acc. %	Test acc. %	Test macro-F1 %	Test ECE %
SmallCNN	100.00	99.60	99.51	11.91
ResNet-18	100.00	99.40	99.01	10.82
DeiT-S	100.00	99.05	98.75	8.71
ResNet-18 (proposed training)	99.99	99.45	99.15	10.83
DeiT-S (proposed training)	99.99	99.50	99.32	9.04
Confidence-weighted ensemble	99.99	99.50	99.32	9.98
Gated fusion (proposed)	99.99	99.50	99.32	9.33

All models are within 0.6 points of each other on clean photos (2,000 test photos: 1 photo = 0.05 points).

5. Is the choice of practice changes fair? (Experiment 4)

The set of practice changes is chosen using only two damage families and then tested on the two others. If the choice only worked on the families it was tuned on, the method would be cheating.

Tuned on	Tested on	Selected op set	Its test acc. %	Best possible	Best acc. %	Baseline %	Gap
noise + blur	weather + digital	PRIME ops only	85.31	ResNet-18 (proposed training)	87.43	80.29	2.12
weather + digital	noise + blur	ResNet-18 (proposed training)	86.04	PRIME ops only	88.78	63.24	2.74

• In both directions the selected set is 2-3 points below the best possible choice, but far above the plain baseline (+5 and +23 points). The changes generalise to damage families they were not chosen on.

• The choice is not stable: tuning on noise + blur picks 'PRIME only', tuning on weather + digital picks the full set. With one seed this may be noise; the full run will tell.

6. Belgian signs (Experiment 5)

Model	Clean %	Clean, shared %	Damaged %	Damaged, shared %	mCE
SmallCNN	95.80	96.55	86.07	87.67	100.0
ResNet-18	94.95	95.90	78.12	80.35	173.5
DeiT-S	95.35	96.10	90.46	92.11	93.2
ResNet-18 (proposed training)	94.05	94.80	88.22	89.46	117.2
DeiT-S (proposed training)	95.60	96.45	91.53	93.02	82.9
Confidence-weighted ensemble	95.75	96.45	91.55	92.60	85.5
Gated fusion (proposed)	**95.85**	**96.60**	**91.73**	**93.00**	**81.7**
v2 stacked gate (FINAL)	**94.20**	**94.85**	**90.52**	**91.73**	**97.3**

2,000 Belgian photos of the 22 sign types that also exist in Germany. 'Shared': the answer may only be one of those 22 types. mCE relative to SmallCNN on Belgian signs.

• The models transfer well: about 94-96% on clean Belgian photos without ever seeing one. The gated fusion is the best on damaged Belgian photos (91.7%, mCE 81.7).

• The proposed training costs ResNet-18 about 1 point on clean Belgian photos (94.95% -> 94.05%) but gains 10 points on damaged ones.

7. Adapting while driving: batch size matters (Experiment 6)



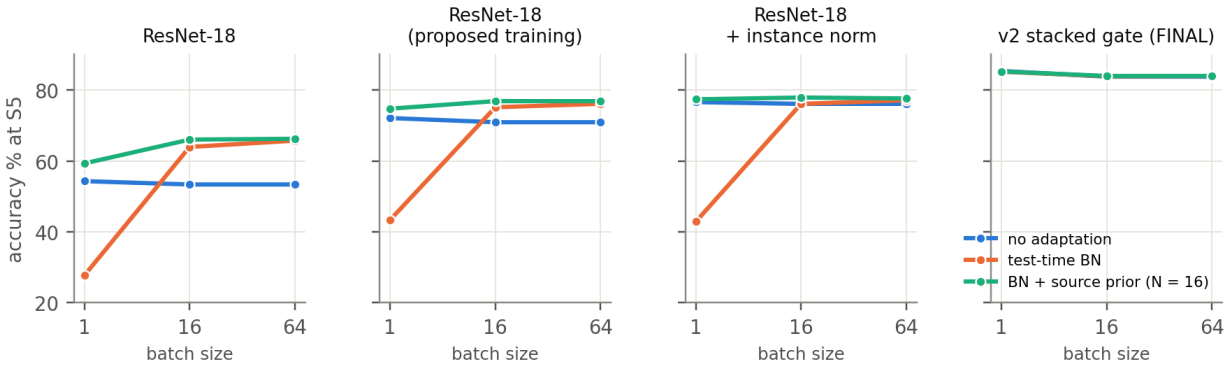
Strongest damage (S5), each damage as its own stream of photos; 500 photos per damage (100 at batch size 1). 'No adaptation' changes slightly at batch size 1 only because fewer photos are used.

• With groups of 16 or more photos, adapting helps the single models a lot: SmallCNN 65% -> 79%, ResNet-18 53% -> 66%, ResNet-18 proposed 71% -> 77%. TENT is only marginally better than just updating the BatchNorm statistics.

• With **one photo at a time** it collapses (ResNet-18 proposed: 72% -> 43% with BN, 29% with TENT). A car that classifies each sign on its own cannot use these methods as they are.

• The fusion barely changes (78.5% -> 78.9%), for the reason given in section 3.

8. Making adaptation work for single photos (Experiment 7, part 2)



Mixing in the training statistics as 16 'pseudo-photos' (source prior) removes the batch-size-1 collapse: ResNet-18 proposed 43% -> 75% at batch 1 (better than no adaptation, 72%), and it is also the best option at batch 16 and 64.

9. The v2 ideas, one change per row (Experiment 7)

Configuration	Val-C (sel.) %	Clean %	Damaged %	S5 %	mCE	Clean ECE %	Damaged ECE %	BTSD damaged %
ResNet-18 (proposed training)	96.01	99.45	86.83	71.94	123.4	10.83	9.82	88.22
ResNet-18 + instance input norm	95.40	99.25	87.96	76.46	117.7	10.23	9.52	87.85
DeiT-S (proposed training)	97.93	99.50	90.34	78.39	78.5	9.04	8.71	91.53
DeiT-S + instance input norm	97.63	99.25	90.30	79.70	87.9	8.98	8.93	90.41
v1 gated fusion	98.03	99.50	90.54	78.75	76.4	9.33	9.11	91.73
v1 average ensemble	97.93	99.50	90.82	79.00	73.5	10.03	11.29	91.57
v1 average + view experts	**98.38**	**99.35**	**92.92**	**83.77**	**64.9**	**10.12**	**12.56**	**91.29**
Average + instance input norm	97.93	99.30	90.93	80.85	79.9	9.70	10.42	90.72
Average + instance norm + view experts	98.03	99.30	92.98	84.81	73.0	9.75	11.73	90.40
v2: + stacked degradation-aware gate	**98.28**	**99.20**	**92.37**	**83.69**	**78.7**	**8.99**	**9.90**	**90.52**
v2 + temperature scaling	**98.28**	**99.20**	**92.37**	**83.69**	**78.7**	**0.48**	**4.11**	**90.52**

Val-C (sel.) is the model-selection score: accuracy under 4 validation damages that are not among the 14 test damages. It is the only column one may use to choose a configuration. Blue row: best Val-C; orange rows: the final architecture.

• **View experts** are the strongest single idea: +2.1 points on damaged photos for the v1 pair (90.82% -> 92.92%) and the best mCE (64.9). They also win on the selection score (98.38%), so choosing by Val-C would have picked the best test model.

• **Instance input normalisation** helps ResNet-18 (+1.1, contrast 91% -> 99%) but not DeiT-S, and costs up to 2 points on clean Belgian photos (ResNet-18 94.05% -> 91.95%).

• **The stacked v2 gate** (92.37%) is not better than simply averaging the view experts (92.92-92.98%), and it is worse on Belgian signs (90.5% vs. 91.7% for the v1 fusion).

• **Temperature scaling** (T = 0.60, fitted on 3,960 held-out validation photos) makes the confidence honest without changing any answer: clean ECE 8.99% -> 0.48%, damaged ECE 9.90% -> 4.11%. T below 1 means the model was under-confident.

10. Training record

Run	Epochs	Best epoch	Best val-C %	Clean val % at best	Minutes	GPU MB
SmallCNN	12	12	93.28	99.80	1.8	1157
ResNet-18	6	4	85.64	99.12	2.0	1494
DeiT-S	6	6	95.69	98.92	3.7	2325
ResNet-18 + mixing	6	5	92.53	99.67	2.3	1495
ResNet-18 + mixing + JSD	6	5	92.48	99.60	5.5	2275
ResNet-18 (proposed training)	6	5	96.02	99.80	5.6	2275
DeiT-S (proposed training)	6	6	97.34	99.85	10.6	5086
Gated fusion (proposed)	2	2	97.44	99.80	1.8	1465
PRIME ops only	6	6	95.11	99.47	5.5	2275
AugMix + PRIME, no geometric ops	6	6	95.84	99.57	5.5	2275
ResNet-18 + instance norm	6	6	95.34	99.57	5.6	2275
DeiT-S + instance norm	6	6	97.59	99.85	10.6	5086
v2 stacked gate (FINAL)	10	2	98.28	99.85	3.3	1160

Total training time 64 min on a T4. Most runs were still improving at their last epoch (best epoch = last epoch), so the shortened pilot schedule probably leaves accuracy on the table; the full run trains 2.5x longer. The fusion v2 gate stopped early at epoch 10 (best at 2); its val-C is measured on the select half of the validation split, so it is not comparable with the other rows.

11. Limits of this pilot

• **One seed.** No spread can be given; the full run uses seeds 0, 1, 2 and reports mean +/- std.

• **2,000 of 12,630 test photos**, and 500 per damage for the adaptation tests (100 at batch size 1).

• **Shortened training** (40% of the epochs).

• **Not comparable with the 10 Oct pilot**: the damage settings were made stronger on 15 Oct, so the same models score lower now.

• **Chinese signs (CTSD)** were not tested (manual download and class-map check still needed).

• Figures of the training notebook (Grad-CAM, Fourier heat map, reliability diagrams) were not in the zip and are not reproduced here.

Suggested next steps

• Run the full profile (PROFILE = '') to confirm the ranking with 3 seeds and the full test set.

• Confirm the final architecture with Val-C (sel.) on the full run: on this pilot Val-C slightly prefers the view-expert average (98.38% vs. 98.28%), a difference too small to decide on one seed.

• Look at fog specifically (for example a dehazing view expert), since nothing tested here improves it.

• Use the source prior for any test-time adaptation, and report batch size 1 explicitly.